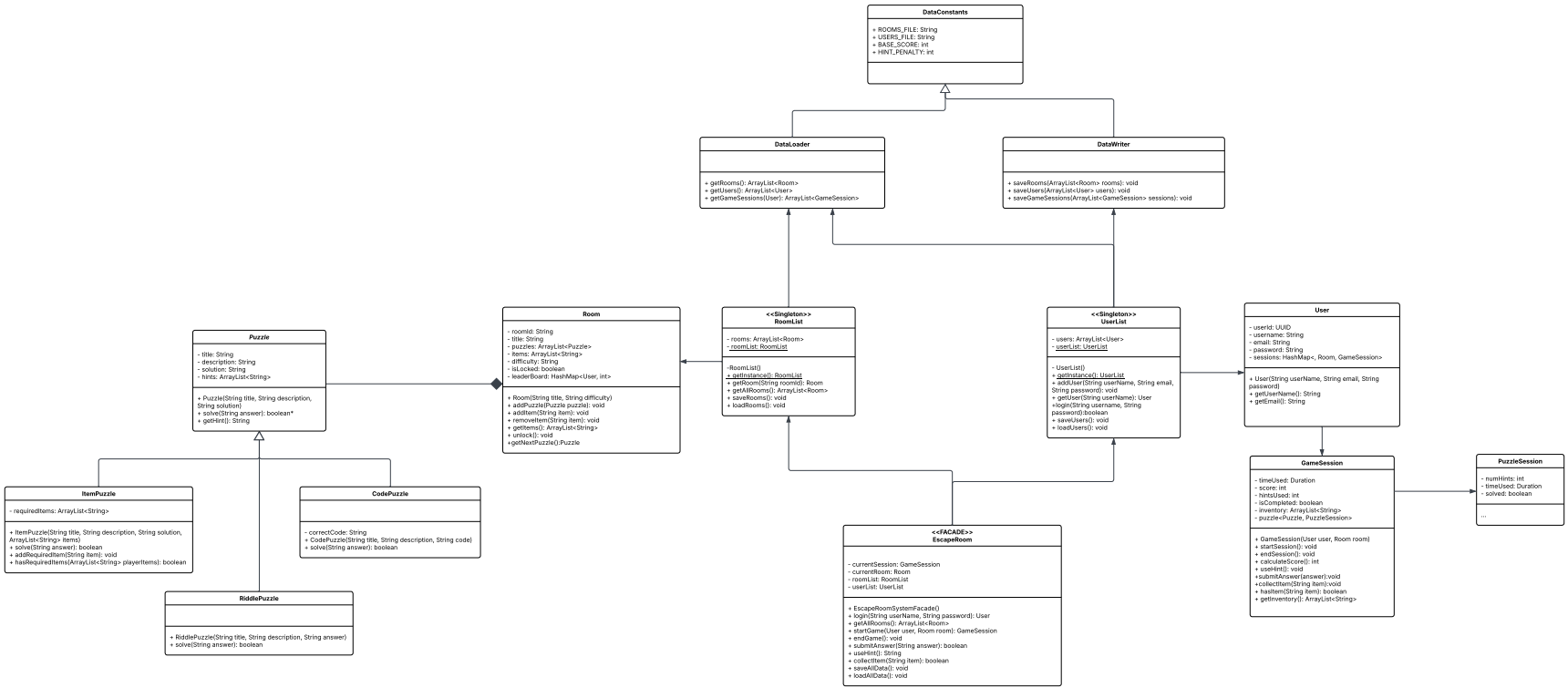

DataConstants
+ ROOMS_FILE: String
+ USERS_FILE: String
+ BASE_SCORE: int
+ HINT_PENALTY: int
DataLoader
+ getRooms(): ArrayList<Room>
+ getUsers(): ArrayList<User>
+ getGameSessions(User): ArrayList<GameSession>
DataWriter
+ saveRooms(ArrayList<Room> rooms): void
+ saveUsers(ArrayList<User> users): void
+ saveGameSessions(ArrayList<GameSession> sessions): void
Puzzle
- title: String
- description: String
- solution: String
- hints: ArrayList<String>
+ Puzzle(String title, String description, String solution)
+ solve(String answer): boolean*
+ getHint(): String
Room
- roomId: String
- title: String
- puzzles: ArrayList<Puzzle>
- items: ArrayList<String>
- difficulty: String
- isLocked: boolean
- leaderBoard: HashMap<User, int>
+ Room(String title, String difficulty)
+ addPuzzle(Puzzle puzzle): void
+ addItem(String item): void
+ removeItem(String item): void
+ getItems(): ArrayList<String>
+ unlock(): void
+getNextPuzzle():Puzzle
<<Singleton>>
RoomList
- rooms: ArrayList<Room>
- roomList: RoomList
-RoomList()
+ getInstance(): RoomList
+ getRoom(String roomId): Room
+ getAllRooms(): ArrayList<Room>
+ saveRooms(): void
+ loadRooms(): void
<<Singleton>>
UserList
- users: ArrayList<User>
- userList: UserList
- UserList()
+ getInstance(): UserList
+ addUser(String userName, String email, String password): void
+ getUser(String userName): User
+login(String username, String password):boolean
+ saveUsers(): void
+ loadUsers(): void
User
- userId: UUID
- username: String
- email: String
- password: String
- sessions: HashMap<, Room, GameSession>
+ User(String userName, String email, String password)
+ getUserName(): String
+ getEmail(): String
ItemPuzzle
- requiredItems: ArrayList<String>
+ ItemPuzzle(String title, String description, String solution, ArrayList<String> items)
+ solve(String answer): boolean
+ addRequiredItem(String item): void
+ hasRequiredItems(ArrayList<String> playerItems): boolean
CodePuzzle
- correctCode: String
+ CodePuzzle(String title, String description, String code)
+ solve(String answer): boolean
RiddlePuzzle
+ RiddlePuzzle(String title, String description, String answer)
+ solve(String answer): boolean
<<FACADE>>
EscapeRoom
- currentSession: GameSession
- currentRoom: Room
- roomList: RoomList
- userList: UserList
+ EscapeRoomSystemFacade()
+ login(String userName, String password): User
+ getAllRooms(): ArrayList<Room>
+ startGame(User user, Room room): GameSession
+ endGame(): void
+ submitAnswer(String answer): boolean
+ useHint(): String
+ collectItem(String item): boolean
+ saveAllData(): void
+ loadAllData(): void
GameSession
- timeUsed: Duration
- score: int
- hintsUsed: int
- isCompleted: boolean
- inventory: ArrayList<String>
- puzzle<Puzzle, PuzzleSession>
+ GameSession(User user, Room room)
+ startSession(): void
+ endSession(): void
+ calculateScore(): int
+ useHint(): void
+submitAnswer(answer):void
+collectItem(String item):void
+ hasItem(String item): boolean
+ getInventory(): ArrayList<String>
PuzzleSession
- numHints: int
- timeUsed: Duration
- solved: boolean
...